

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - GPS L5 (1176 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Mark Sullivan
Reviewer:	Dr. Raj Patel
Design Center:	Osaka Design Center
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting GPS L5 (1176 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 5054.8 MHz and bandwidth of 386.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: GPS L5 (1176 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: 36° YX LiTaO3
- Package: WLP (Wafer Level Package)
- Design Tool: Synopsys CustomSim
- Design Center: Osaka Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting GPS L5 (1176 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5054.8 MHz
Bandwidth (3 dB):	386.3 MHz
Insertion Loss Target:	<= 3.8 dB
Return Loss Target:	>= 9.0 dB
Out-of-Band Rejection:	>= 25.0 dB
Isolation (Duplexer):	>= 24.0 dB
Q Factor:	8448
Temperature Coefficient:	-22.5 ppm/°C
Die Size:	2.0 x 1.3 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 5054.8 MHz, BW 386.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15164 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.8 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 579 MHz offset >= 25.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 14.2 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-03-08
Devices Tested: 12
Insertion Loss (meas): 3.48 dB
Return Loss (meas): 10.8 dB
Rejection (meas): 28.1 dB
Isolation (meas): 29.2 dB
Q Factor (meas): 8448
Group Delay Variation: 2.7 ns
Power Handling: +36 dBm
Die Yield: 87.8%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated below-target performance for GPS L5 (1176 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

Author:	Mark Sullivan	Date:	2024-03-08
Reviewer:	Dr. Raj Patel	Date:	2024-03-12
Approver:	Dr. Hans Weber	Date:	2024-03-16

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